- Functional Requirements Document (FRD) for Data Processing and Loading**

===

- Introduction**

The purpose of this document is to outline the functional requirements for the data processing and loading component of the project. This component is responsible for reading data from CSV files, loading it into Delta tables, transforming and aggregating the data, and loading it into the `ordersummary` and `customeraggregatespend` tables.

- Functional Requirements**

1. Data Ingestion

- The system shall read CSV data from the specified volumes: `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata` and `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata`.
- The system shall load the data into Delta tables `customer` and `order`.

2. Data Quality

- The system shall remove null and duplicate records from both `customer` and `order` tables.

3. Data Transformation and Loading

- The system shall create an `ordersummary` table if it does not exist in the catalog `gen_ai_poc_databrickscoe` and schema `sdlc_wizard`.
- The system shall join the `customer` and `order` data using the `CustId` field and load the data into an SCD Type 2 table `ordersummary`.

4. SCD Type 2 Implementation

- The system shall implement SCD Type 2 logic to update the `ordersummary` table whenever there is a change in the `customer` table.
- Old records shall be marked as inactive, and new records shall be marked as active, with corresponding changes to the `StartDate` and `EndDate` fields.

5. Data Aggregation

- The system shall create a new table `customeraggregatespend` with columns `Name`, `TotalAmount`, and `Date` in the catalog `gen_ai_poc_databrickscoe` and schema `sdlc_wizard`, if it does not exist.
- The system shall aggregate the `TotalAmount` column from the `ordersummary` table and group it by the `Name` and `Date` columns.
- The system shall load the aggregated data into the `customeraggregatespend` table.

- Data Flow Diagram**

- Data Processing and Loading Process**

The data processing and loading process shall involve the following steps:

1. Read CSV data from the specified volumes.

2. Load the data into Delta tables `customer` and `order`.

3. Remove null and duplicate records from both tables.

4. Join the `customer` and `order` data using the `CustId` field.

5. Load the joined data into the `ordersummary` table.

6. Implement SCD Type 2 logic to update the `ordersummary` table.

7. Aggregate the `TotalAmount` column from the `ordersummary` table.

8. Load the aggregated data into the `customeraggregatespend` table.

- Interface Requirements**

- The system shall be able to read CSV data from the specified volumes.
- The system shall be able to load data into Delta tables `customer` and `order`.
- The system shall be able to create the `ordersummary` and `customeraggregatespend` tables if they do not exist.

- Data Requirements**

- The schema of the `customer` table is: `CustId`, `Name`, `EmailId`, `Region`.
- The schema of the `order` table is: `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId`.
- The schema of the `ordersummary` table is: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`.
- The schema of the `customeraggregatespend` table is: `Name`, `TotalAmount`, `Date`.

- Error Handling and Logging**

- The system shall log any errors that occur during the data processing and loading process.
- The system shall be able to handle large volumes of data efficiently.

By following this FRD, the development team should be able to design and implement a data processing and loading component that meets the business requirements and ensures data consistency and integrity.